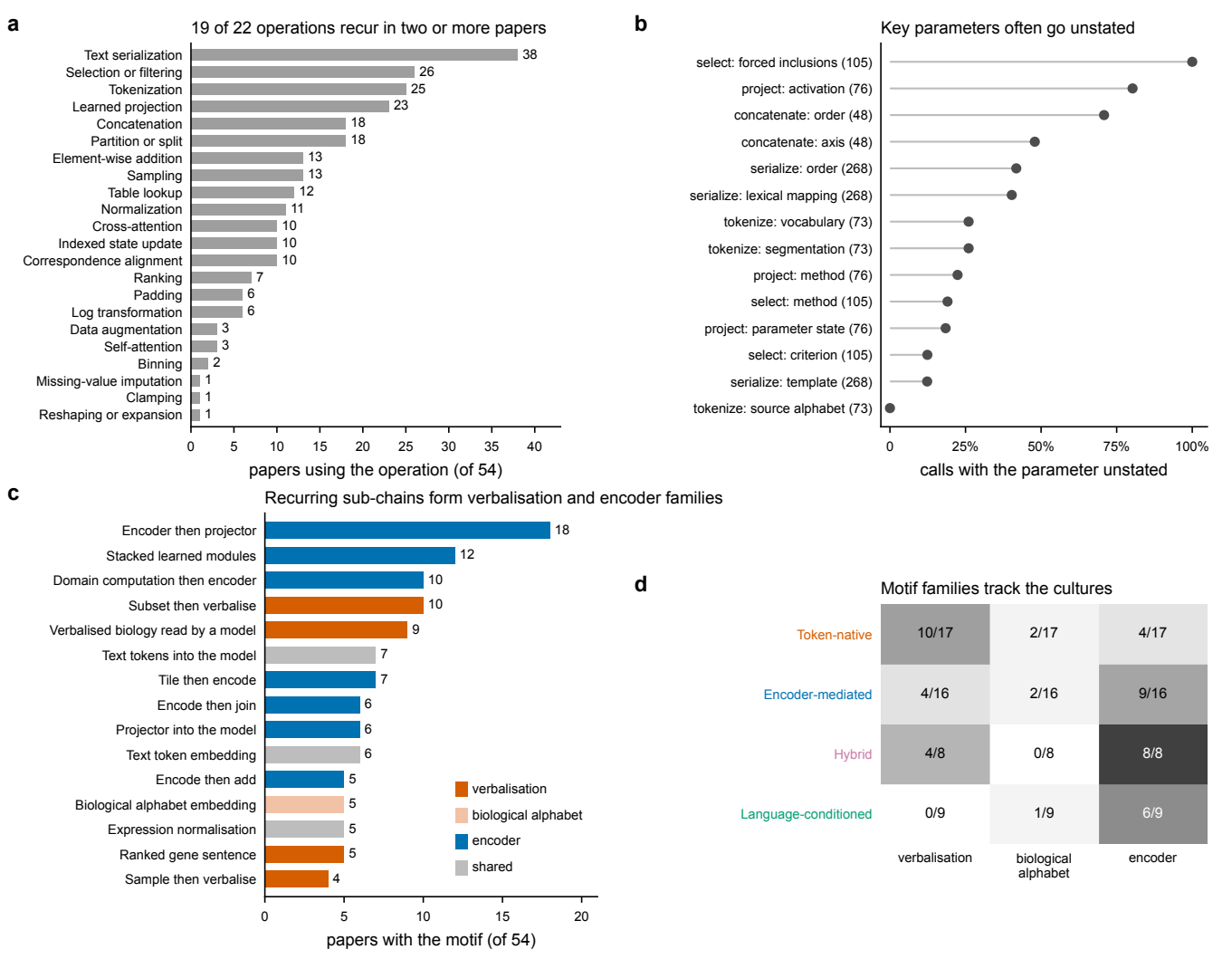
a
19 of 22 operations recur in two or more papers
Text serialization 38
Selection or filtering 26
Tokenization 25
Learned projection 23
Concatenation 18
Partition or split 18
Element-wise addition 13
Sampling 13
Table lookup 12
Normalization 11
Cross-attention 10
Indexed state update 10
Correspondence alignment 10
Ranking 7
Padding 6
Log transformation 6
Data augmentation 3
Self-attention 3
Binning 2
Missing-value imputation 1
Clamping 1
Reshaping or expansion 1
0 5 10 15 20 25 30 35 40
papers using the operation (of 54)
b
Key parameters often go unstated
select: forced inclusions (105)
project: activation (76)
concatenate: order (48)
concatenate: axis (48)
serialize: order (268)
serialize: lexical mapping (268)
tokenize: vocabulary (73)
tokenize: segmentation (73)
project: method (76)
select: method (105)
project: parameter state (76)
select: criterion (105)
serialize: template (268)
tokenize: source alphabet (73)
0 25% 50% 75% 100%
calls with the parameter unstated
c
Recurring sub-chains form verbalisation and encoder families
Encoder then projector 18
Stacked learned modules 12
Domain computation then encoder 10
Subset then verbalise 10
Verbalised biology read by a model 9
Text tokens into the model 7
Tile then encode 7
Encode then join 6
Projector into the model 6
Text token embedding 6
Encode then add 5
Biological alphabet embedding 5
Expression normalisation 5
Ranked gene sentence 5
Sample then verbalise 4
verbalisation
biological alphabet
encoder
shared
0 5 10 15 20
papers with the motif (of 54)
d
Motif families track the cultures
Token-native 10/17 2/17 4/17
Encoder-mediated 4/16 2/16 9/16
Hybrid 4/8 0/8 8/8
Language-conditioned 0/9 1/9 6/9
verbalisation
biological alphabet
encoder